WATERWATCH · DESIGN DECISION RECORD

WaterWatch recommendations accepted

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DOCUMENT CONTROL

Document ID	WWT-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/waterwatch		
Source	docs/decisions/0002-recommendations-accepted.md @ 96e344a		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	O1 to O3 recorded as decided by Amish on 2026-10-02
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted for items D11 to D14; items O1 to O3 decided by Amish on 2026-10-02 (WWT-DEC-001)

Context

After the TRL 3 session, `docs/REVIEW.md` (session 2026-09-25, TRL 3) listed two new items awaiting Amish, each with a recommendation (the R1 target and the pH probe trigger), and two suggestions offered with a recommendation to adopt them (a 15 min confirming reading and casting the footing on a survey visit). On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every item that carried a recommendation is therefore decided in favor of that recommendation; where several options were offered, the recommended option is the decision. Items with no recommendation stay open. The items D1 to D10 in WWT-DDR-001 were already decided and are unchanged. TRL 4 remains on hold by Amish's instruction, and nothing here authorizes building, testing or purchasing.

Decision

Table 1. Newly decided items and what changed in the repo.

#	Item	Decision	What changed
D11	R1 free chlorine target	Option (b): relax R1 from ±0.1 mg/L below 1.0 mg/L and ±15 % above to ±0.2 mg/L or ±25 %, whichever is greater. Decided by Amish, 2026-09-25: go with recommendation.	WWT-REQ-001 v0.4, R1 target restated. WWT-CAL-001 v0.2, section E: cases meeting R1 at zero drift rise from 2 of 8 to 8 of 8 (with D12); room for drift from 0.038 to 0.130 mg/L. R1 moves from not met to at risk; R10 from not met to at risk (drift allowance about 0.13 mg/L per month, was 0.04).
D12	pH probe trigger	Fit the pH probe variant at sites whose pH is above 7.5 or moves more than 0.2 between visits; site pH (WWT-DDR-001, D2) elsewhere. Decided by Amish, 2026-09-25: go with recommendation.	Design change: the cell lid gains a spare 12 mm port with a blanking plug (`cad/src/model.py` , STEP and STL re-exported; WWT-DWG-001 Rev P1 to P2; BOM item 9 $12 to $13, total $282 to $283). WWT-REQ-001 v0.4, R12 restated: the pH probe (about $60 to $100, estimate) is priced per variant site, outside the $300 base-unit budget; a variant unit costs about $343 to $383. `budget_usd` stays at $300.
D13	Confirming reading	Firmware rule: after a first threshold crossing, take the confirming reading 15 min later instead of at the next hour. Decided by Amish, 2026-09-25: go with recommendation.	WWT-REQ-001 v0.4, R5 restated; WWT-PRC-001 v0.4, How it works. WWT-CAL-001 v0.2, H2 and H3: worst case from event to SMS 122 min to 77 min, at 9.7 mWh and 0.75 L per event. Recorded as a rule; firmware beyond a sketch is TRL 4 work and is on hold.
D14	Footing on the survey visit	Dig and cast the pole footing, with the pole set, on the survey visit before installation day. Decided by Amish, 2026-09-25: go with recommendation.	WWT-REQ-001 v0.4, R11 restated as 2 h on installation day. WWT-CAL-001 v0.2, J2: installation day critical path 140 min to 65 min. R11 moves from at risk to met on paper.

No pitch, problem or `budget_usd` change was recommended, so `project.yaml` is unchanged; `trl` and `trl_target` stay at 3.

Table 2. Items left open on 2026-09-25, decided by Amish on 2026-10-02 (WWT-DEC-001).

#	Item	Status
O1	First co-design partner and region (WWT-DDR-001, O1). Community designs pick co-design partners per area later.	Decided 2026-10-02: Water Mission as the first candidate to approach, region from its East African country programs
O2	Alert language and recipients beyond the R13 rule; no recommendation was made.	Decided 2026-10-02: SMS in the local language with an English copy to the operator, to the caretaker and the operator's maintenance contact, weekly summary to the district water officer
O3	Data ownership and access; no recommendation was made.	Decided 2026-10-02: owned by the scheme operator, visible to the water committee and district offices, published only with the operator's written agreement

Consequences

- Requirement status (WWT-CAL-001 v0.2): 0 not met, 4 at risk (R1, R10, R2, R9), 6 met on paper, 3 met by design. Before: 2 not met, 3 at risk, 5 met on paper, 3 met by design.
- At 40 °C, 1.5 mg/L free chlorine at pH 7.5 with site pH misses the relaxed R1 by 0.006 mg/L (WWT-CAL-001, E7). Warm-water sites near pH 7.5 are candidates for the pH probe under the same rule.
- Controlled documents revised: WWT-PRC-001 v0.4, WWT-REQ-001 v0.4, WWT-CAL-001 v0.2, WWT-DDR-001 v0.2; drawing WWT-DWG-001 Rev P2; concept media regenerated.
- No cross-repo action arises from these items.
- R1 and R10 now hinge on the chlorine sensor's drift, which only a bench test can settle. That is TRL 4 work and is on hold.